

Agile Project Documentation

(For MCA Semester II — Agile CI-Based Project)

Project Title

TB Real Estate Platform – Property Listing & Management System

Project Status: **Completed**

Last Updated: March 2026

1. Project Overview

1.1 Team Members

Name	Role
Thejazo Tachu	Scrum Master, Frontend Developer
Bageesh Kumar Sharma	Product Owner, Backend Developer, Tester

1.2 Abstract

TB Real Estate Platform is a modern, web-based real estate property listing and management system designed to simplify the interaction between home buyers and property owners. The platform enables users to explore residential properties such as 1BHK to 4BHK apartments, villas, and duplex homes across multiple Indian states.

The system provides advanced property search and filtering, detailed property pages with image galleries, user authentication, property comparison, inquiry handling, visit scheduling, and feedback mechanisms.

The project was developed using the Agile methodology, ensuring incremental delivery, adaptability to change, and continuous feedback. The frontend is implemented using React 18 with TypeScript, while the backend leverages PostgreSQL with Row Level Security (RLS) for secure data access.

Current Status: The project has been successfully completed across 5 development sprints. All core features, user interaction modules, email notification systems, and documentation have been finalized and delivered.

2. Technology Stack (Implemented)

2.1 Frontend Technologies

Technology	Purpose	Status
React 18	UI Development	Implemented
TypeScript	Type Safety	Implemented
Vite	Build Tool	Implemented
Tailwind CSS	Styling	Implemented
shadcn/ui	UI Components	Implemented
React Router DOM v6	Routing	Implemented
TanStack React Query	Data Fetching	Implemented
React Hook Form + Zod	Form Validation	Implemented
Lucide React	Icons	Implemented

2.2 Backend Technologies

Technology	Purpose	Status
PostgreSQL	Database	Implemented
Authentication Service	User Login/Signup	Implemented
Row Level Security	Data Protection	Implemented
Edge Functions	Serverless Logic	Implemented
Resend API	Email Notifications	Implemented

3. System Architecture

The system follows a three-tier architecture consisting of:

- Client Layer: Web browsers (Desktop, Tablet, Mobile)
- Application Layer: React-based frontend with TypeScript
- Data Layer: PostgreSQL database with RLS policies

This architecture ensures scalability, maintainability, and secure data access.

4. Database Design

4.1 Table Specifications

Table	Purpose	RLS Policies	Status
profiles	User profiles	User-specific access	Active
property_listings	Property data	Owner + approved public	Active
saved_properties	User favorites	User-specific	Active
scheduled_visits	Visit bookings	User-specific	Active
inquiries	Property inquiries	Submitter + Owner view	Active
feedback	User reviews	Approved public	Active

5. Problem Statement

Traditional real estate platforms suffer from:

- Poor user experience due to cluttered interfaces
- Limited property filtering and comparison capabilities
- Lack of visit scheduling and inquiry management
- Absence of personalized user accounts
- No communication channel between buyers and owners

6. Proposed Solution

The TB Real Estate Platform addresses these challenges by offering:

- Advanced search and filtering (location, price, BHK type, state)
- Responsive, modern UI accessible on desktop, tablet, and mobile
- Secure user authentication with email/password
- Property comparison for informed decision-making
- Visit scheduling with email confirmation notifications
- Inquiry handling and feedback systems

7. Agile Methodology

7.1 Agile Practices Followed

- Sprint-based development (5 sprints across 5 weeks)
- User stories with acceptance criteria
- Incremental feature delivery
- Continuous testing during development
- Regular sprint reviews and retrospectives

7.2 Team Roles & Responsibilities

Role	Member	Responsibilities
Product Owner	Bageesh Kumar Sharma	Define requirements, manage backlog, accept stories
Scrum Master	Thejazo Tachu	Facilitate Scrum, remove blockers, ensure practices
Frontend Dev	Both	UI/UX, React components, responsive design
Backend Dev	Bageesh Kumar Sharma	Database design, RLS policies, API integration
Tester	Bageesh Kumar Sharma	Test cases, bug tracking, UAT
DevOps	Both	Deployment, version control

8. Product Backlog Summary

Epic	Status
Core Property Browsing	Completed
User Authentication	Completed
Favorites & Visits	Completed
Feedback & Inquiry	Completed
Property Owner Listing	Completed
Email Notifications	Completed
Property Comparison	Completed

Before development began, the team conducted detailed discussions to understand real-world real estate challenges, user expectations, and overall project feasibility. Based on these discussions, the product backlog was created and prioritized.

The Core Property Browsing module was identified as the highest priority and has been fully implemented and tested. User Authentication was treated as a critical requirement and includes secure login, registration, and session management, all of which have been successfully validated. The Favorites and Visit Scheduling features were developed to allow users to save properties and schedule visits. The Feedback and Inquiry module enables effective communication between users and property owners. The Email Notification system was integrated using Resend API via Edge Functions. The Property Comparison tool enables users to compare up to 3 properties side-by-side.

9. Sprint Progress

Sprint 1 (Week 1): Foundation Setup — COMPLETED

Previous Sprint Reference:

This was the initial sprint of the project. Prior to development, requirement analysis was conducted, the technology stack was finalized, and the development environment was prepared.

Tasks Completed in This Sprint:

1. Project initialization using Vite, React 18, and TypeScript
2. Installation and configuration of Tailwind CSS and shadcn/ui
3. Setup of React Router DOM v6 for client-side routing
4. Development of Header component with navigation links
5. Development of Footer component with contact and social links
6. Creation of Home page with hero section and featured properties
7. Development of About page with company and team information
8. Implementation of Contact page with inquiry form structure
9. Configuration of responsive layout for mobile, tablet, and desktop

Role-wise Contribution:

Thejazo Tachu (Scrum Master, Frontend Developer):

- Facilitated sprint planning and daily stand-up coordination
- Set up frontend project structure and routing
- Developed Header and Footer components with responsive design
- Implemented Home page layout with hero section and navigation flow
- Ensured mobile-first responsive design across all pages

Bageesh Kumar Sharma (Product Owner, Backend Developer, Tester):

- Defined user stories and acceptance criteria for Sprint 1
- Assisted in repository setup and version control workflow
- Developed About page with company and team information content
- Designed Contact page structure and form layout
- Performed initial testing of routing and navigation

Sprint Outcome:

A stable project foundation was established with responsive navigation, three core pages (Home, About, Contact), and a scalable frontend component architecture.

Sprint 2 (Week 2): Property Features — COMPLETED

Previous Sprint Reference:

Building upon the foundation established in Sprint 1, the team proceeded to implement the core property listing functionality. The routing and UI foundation enabled seamless integration of property-related features.

Tasks Completed in This Sprint:

10. Creation of comprehensive property data structure with 45+ properties across 12 Indian states
11. Development of Properties listing page with grid layout
12. Implementation of advanced filtering system (location, price range, BHK type, state)
13. Creation of reusable PropertyCard component with image, specifications, and pricing
14. Development of PropertyDetail page with image gallery
15. Implementation of property specifications display (area, bedrooms, bathrooms, amenities)
16. Addition of property type categories (1BHK, 2BHK, 3BHK, 4BHK, Villa, Duplex)
17. Integration of search functionality with real-time filtering
18. Development of Services page with company offerings

Role-wise Contribution:

Thejazo Tachu (Scrum Master, Frontend Developer):

- Coordinated sprint review sessions and backlog refinement
- Developed Properties listing page UI with filter sidebar
- Implemented advanced filtering and search logic
- Created PropertyCard and image gallery components
- Maintained consistent UI styling using Tailwind CSS design tokens

Bageesh Kumar Sharma (Product Owner, Backend Developer, Tester):

- Curated and validated property data for 45+ listings across multiple states
- Designed property data schema with comprehensive attributes
- Developed PropertyDetail page with full specifications display
- Created Services page with service offerings
- Conducted functional testing of filter combinations and listing accuracy

Sprint Outcome:

A fully functional property browsing module with advanced filtering, detailed property views with image galleries, and comprehensive property information was delivered.

Sprint 3 (Week 3): User Authentication & Interaction — COMPLETED

Previous Sprint Reference:

With property browsing features completed in Sprint 2, the focus shifted to implementing user authentication and personalized interaction features.

Tasks Completed in This Sprint:

19. Integration of backend database (PostgreSQL) with Row Level Security
20. Implementation of user signup and login functionality (email/password)
21. Creation of user profiles table and related database tables
22. Development of saved properties feature with database persistence
23. Implementation of visit scheduling functionality with date and time selection
24. Development of inquiry submission system with database storage
25. Creation of feedback and rating system
26. Development of My Visits page for viewing scheduled appointments
27. Development of Saved Properties page for viewing favorites

Role-wise Contribution:

Thejazo Tachu (Scrum Master, Frontend Developer):

- Coordinated sprint activities and resolved blockers
- Developed authentication UI components (login/signup forms)
- Created ScheduleVisitModal component with form validation
- Implemented UserMenu component with dropdown navigation
- Developed My Visits and Saved Properties pages
- Integrated UI feedback and toast notifications

Bageesh Kumar Sharma (Product Owner, Backend Developer, Tester):

- Designed database schema and implemented RLS policies for all tables
- Integrated backend authentication services with email/password
- Developed AuthContext for global authentication state management
- Implemented CRUD operations for user features
- Conducted end-to-end testing of authentication flows
- Validated data security and access control through RLS enforcement

Sprint Outcome:

Secure user authentication and personalized user features were successfully implemented, enabling meaningful user interaction with saved properties, visit scheduling, inquiry submission, and feedback mechanisms.

Sprint 4 (Week 4): Enhancement, Optimization & Documentation — COMPLETED

Previous Sprint Reference:

After stabilizing user authentication and interaction features in Sprint 3, this sprint focused on system enhancement, optimization, email integration, and academic documentation.

Tasks Completed in This Sprint:

28. Development of property comparison functionality for side-by-side analysis
29. Implementation of email notification system using Resend API via Edge Functions
30. Integration of welcome email on user registration
31. Implementation of visit confirmation and inquiry acknowledgment emails
32. UI/UX improvements and responsive design refinements
33. Performance optimization and code refactoring
34. Expansion of property listings with additional properties per state
35. Cross-browser compatibility testing
36. Preparation of ER and DFD diagrams
37. Updating Agile project documentation

Role-wise Contribution:

Thejazo Tachu (Scrum Master, Frontend Developer):

- Coordinated sprint execution and documentation reviews
- Developed PropertyComparison component with comparison modal
- Enhanced frontend performance and responsiveness
- Refined search bar functionality on the home page
- Conducted cross-browser UI testing on Chrome, Firefox, and Safari
- Updated frontend-related documentation

Bageesh Kumar Sharma (Product Owner, Backend Developer, Tester):

- Implemented send-email Edge Function with Resend API integration
- Developed email templates for welcome, visit, inquiry, and feedback notifications
- Configured RESEND_API_KEY secret for email service
- Expanded and validated property data across states
- Prepared ER diagrams and DFD diagrams
- Conducted integration and system testing of email notification system
- Updated Agile documentation and reports

Sprint Outcome:

The system reached a stable and optimized state with enhanced features including property comparison, email notifications, expanded listings, and comprehensive academic documentation.

Sprint 5 (Week 5): Final Integration, Testing & Project Completion — COMPLETED

Previous Sprint Reference:

Upon completion of Sprint 4's enhancement and documentation phase, the final sprint focused on completing all remaining functionalities, comprehensive testing, final documentation, and project delivery.

Tasks Completed in This Sprint:

38. Final integration testing of all modules (authentication, properties, visits, inquiries, feedback)
39. End-to-end validation of email notification system across all triggers
40. Functional search bar integration on the home page with URL parameter synchronization
41. View Details navigation from featured property cards on the home page
42. Final responsive design testing across mobile, tablet, and desktop viewports
43. Security review of all RLS policies and authentication flows
44. Final code refactoring and performance optimization
45. Completion and finalization of Agile project documentation
46. Completion of project thesis documentation
47. Final User Acceptance Testing (UAT) for all features

Role-wise Contribution:

Thejazo Tachu (Scrum Master, Frontend Developer):

- Facilitated final sprint ceremonies and project completion review
- Implemented functional search bar on the home page with property navigation
- Added View Details links from featured property cards to detail pages
- Conducted final responsive design validation across all breakpoints
- Performed cross-browser testing and resolved remaining UI issues
- Prepared final frontend documentation and component specifications
- Coordinated project delivery and presentation preparation

Bageesh Kumar Sharma (Product Owner, Backend Developer, Tester):

- Conducted comprehensive end-to-end testing of all system modules
- Validated email notification delivery for all trigger events
- Performed final security audit of RLS policies and data access controls
- Verified database integrity and data consistency across all tables
- Completed final UAT and documented test results
- Finalized Agile documentation with Sprint 5 completion details
- Prepared project deliverables for academic submission

Sprint Outcome:

The project was successfully completed with all planned features implemented, tested, and documented. The platform is fully functional and ready for academic evaluation and submission.

10. Testing

Testing was performed continuously throughout the development process to ensure system reliability and correctness.

TC ID	Module	Description	Expected Result	Status
TC01	Properties	Filter by location	Filtered results displayed	Pass
TC02	Properties	Filter by BHK type	Correct listings shown	Pass
TC03	Properties	View property details	Full details shown	Pass
TC04	Properties	Search from home page	Navigates to filtered results	Pass
TC05	Auth	User registration	Account created successfully	Pass
TC06	Auth	User login	Session established	Pass
TC07	Auth	Welcome email sent	Email received on signup	Pass
TC08	Favorites	Save property	Added to favorites	Pass
TC09	Visits	Schedule visit	Visit recorded + email sent	Pass
TC10	Inquiry	Submit inquiry	Inquiry saved + email sent	Pass
TC11	Feedback	Submit feedback	Feedback recorded	Pass
TC12	Compare	Compare properties	Side-by-side comparison view	Pass
TC13	Navigation	View Details from home	Navigates to property detail	Pass
TC14	Responsive	Mobile layout	Properly responsive	Pass

11. Current Project Status

Module	Progress	Status
Frontend UI	100%	Complete
Authentication	100%	Complete
Property Browsing	100%	Complete
Property Details	100%	Complete
Search & Filter	100%	Complete
User Favorites	100%	Complete
Visit Scheduling	100%	Complete
Inquiry System	100%	Complete
Feedback System	100%	Complete
Property Comparison	100%	Complete
Property Listing (Owner)	100%	Complete
Email Notifications	100%	Complete

Overall Project Completion: 100%

The TB Real Estate Platform has been fully completed. All frontend development work has been successfully finished, and the application interface is stable, responsive, and user-friendly. Backend services have been implemented including database operations with Row Level Security, user authentication, and email notifications via Edge Functions.

12. Future Enhancements

While the core project has been completed, the following enhancements are identified for future development:

- Admin Dashboard: To manage users, properties, and system activities
- Image Upload: Allow property owners to upload images directly
- EMI/Mortgage Calculator: Financial planning tool for buyers
- Map-based Property View: Interactive map integration
- Mobile Application: React Native app for mobile access
- Chat Support: Real-time messaging between buyers and owners

13. Deliverables

Deliverable	Status
Functional real estate web application	Delivered
Source code repository with version control	Delivered
Database schema with RLS policies	Delivered
Live deployment	Delivered
ER and DFD diagrams	Delivered
Agile project documentation	Delivered
Technical thesis documentation	Delivered

14. Conclusion

The TB Real Estate Platform demonstrates the effective use of Agile methodology in building a scalable, user-centric real estate application. Incremental development, continuous testing, and sprint-based planning enabled the successful implementation of all planned features across 5 development sprints.

Key achievements include:

- Fully functional property browsing and discovery with 45+ listings
- Secure authentication with email/password and session management
- Complete user interaction features (favorites, visits, inquiries, feedback)
- Email notification system with automated welcome, visit, and inquiry emails
- Property comparison tool for informed decision-making
- Responsive design across desktop, tablet, and mobile devices
- Database security with comprehensive RLS policies

The project has been successfully completed and is ready for academic evaluation and submission.

Project Status: Completed